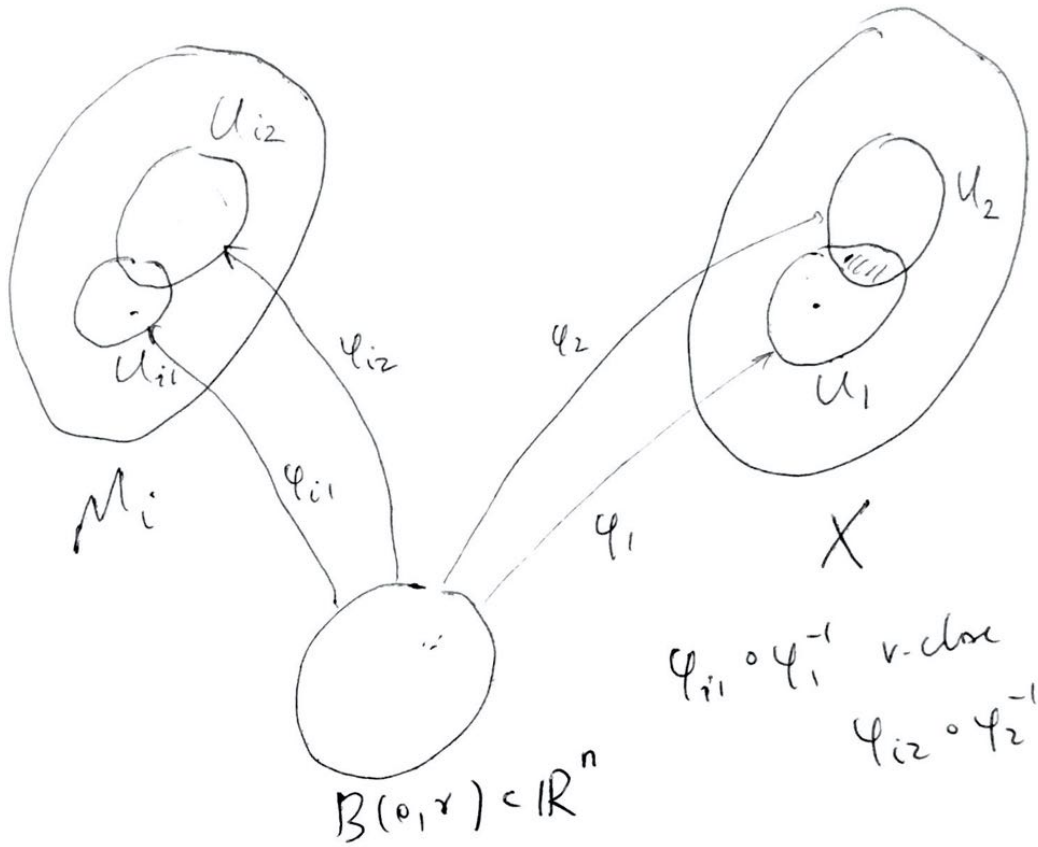
<u>Step 3</u> <u>Claim</u>: $(M_i, g_i, p_i) \longrightarrow (X, g, p)$ in $C^{m,\beta}$.



Key observation is that

$$\varphi_{i1} \circ \varphi_1^{-1} \quad \text{and} \quad \varphi_{i2} \circ \varphi_2^{-1}$$

"converge to each other" on $U_1 \cap U_2$ as $i \to \infty$

idea is to define a map on $U_1 \cap U_2$ by interpolating between $\varphi_{i1} \circ \varphi_1^{-1}$ and $\varphi_{i2} \circ \varphi_2^{-1}$ using a partition of unity. $\square$

<u>Corollary</u> Given n, Q, r, m, α, D, V the set of $(M, g, p) \in \mathcal{M}^{m,\alpha}(n, Q, r)$ which also satisfy either $\operatorname{diam} < D$ or $\operatorname{vol} < V$ has finitely many diffeomorphism types.

<u>Q</u> Can we find bounds $\|(M,g)\|_{C^{m,\alpha},r} \leq Q$ in terms of geometric bounds on (M,g)?

<u>Lemma</u> Let (M,g) be Riemannian,

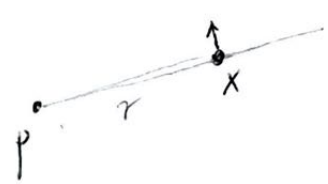
injrad $\geq i_0$ (radius of ball on which exponential map is a diffeo.)

and $|Rm| \leq K$, $p \in M$, $r(x) = d(p,x)$, $r : M \to \mathbb{R}$

The Hessian H of r ($H(v) = \nabla_v \operatorname{grad} r$ for $v \in T_xM$)

has $\operatorname{grad} r$ as an eigenvector with eval 0 and other eigenvalues are in interval

$$\left[\sqrt{K}\cot(\sqrt{K}\,r),\ \sqrt{K}\coth(\sqrt{K}\,r)\right]$$



<u>Proof</u> Riccati equation

$$\frac{\partial H}{\partial r} + H^2 = -R \qquad \left(R(v) = Rm(v, \operatorname{grad} r) \cdot \operatorname{grad} r \right)$$

R has eigenvalues in $[-K, K]$

eigenvalues of H satisfy

$$-K \le \lambda'(r) + \lambda^2(r) \le K \qquad \square$$

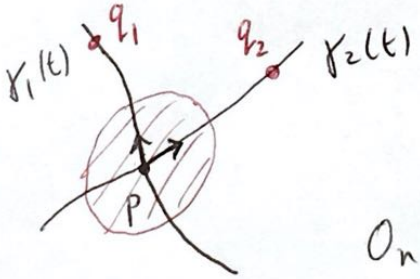
<u>Prop</u> If $|Rm| \le K$, injrad $\geq i_0$ then $\|(M, g)\|_{C^{0,\alpha}, r} \le Q$ for some r, Q dep. on n, K, r_0.

<u>Proof</u> Let $p \in M$, we need to construct a "good" coordinate chart at p.

Let $e_1, \dots, e_n \in T_pM$ orthonormal

$\gamma_i(t) =$ geodesic with $\gamma_i(0) = p$, $\dot{\gamma}_i(0) = e_i$

$$q_i = \gamma_i\left(\frac{i_0}{4\sqrt{K}}\right)$$



$$r_i(x) = d(q_i, x)$$

On $B\left(p, \frac{i_0}{8\sqrt{K}}\right)$ the r_i have uniform Hessian bounds

Let $\varphi_p(x) = (r_1(x), \dots, r_n(x))$

$$\varphi_p : B\left(p, \frac{i_0}{8\sqrt{K}}\right) \longrightarrow \mathbb{R}^n$$

Let $g_p = \varphi_p^* g_{Euc}$. Then $g_p(p) = g(p)$

$\Longrightarrow$ using Hessian bounds we get

$|d\varphi_p|, |d\varphi_p^{-1}| \leq e^Q$ on $B(p, \varepsilon)$

for ε, Q dep. on i_0, K

$\rightsquigarrow \|(M, g, p)\|_{C^{0,\alpha}, r} \leq Q$ for some r, Q dep. on K, i_0 $\square$

<u>Lemma</u> Suppose (M, g) satisfies $|Rm| \leq K$, $\mathrm{Vol}\, B(p,1) \geq v > 0$ for all $p \in M$. Then injrad $M \geq i_0$ for i_0 dep. on n, K, v.

<u>Proof</u> Argue by contradiction. Suppose (M_i, g_i) satisfy $|Rm| \leq K$, $\mathrm{vol}(B(p,1)) \geq v$ but injrad $M_i \to 0$. $\exists\, p_i \in M_i$ s.t.

injrad $M_i \leq$ injrad $p_i \leq 2$ injrad $M_i = 2\delta_i \to 0$

Consider rescaled manifolds $(\tilde{M}_i, \tilde{g}_i, \tilde{p}_i)$

$\tilde{M}_i = M_i$, $\tilde{g}_i = \delta_i^{-2} g_i$, $\tilde{p}_i = p_i$

now $\quad \text{injrad } \tilde{p}_i = \delta_i^{-1} \text{ injrad } p_i \in [1,2]$

and $\quad \text{injrad } \tilde{M}_i = 1$

$$|Rm\, \tilde{M}_i| \leq \delta_i K \to 0$$

in M_i we had $\quad Vol(B(p_i, 1)) \geq v$

relative vol. comparison $\Rightarrow Vol(B(p_i, r)) \geq v' r^n$
$\forall r < 1$

so in $\tilde{M}_i \quad Vol(B(\tilde{p}_i, r)) \geq v' r^n$
for all $r < \delta_i^{-1}$

Previous result $\Rightarrow$ can extract pointed
$C^{0,\beta}$ limit $(\tilde{M}, \tilde{g}, \tilde{p})$ with
$Rm\, \tilde{M} = 0$

(use Cheeger - Gromoll splitting)
to see that $\tilde{M}$ must be $\mathbb{R}^n$